Research audit for reviewing ECA-UWB manuscript

ECA-UWB claims a lightweight dual-branch network (2,374 parameters) for UWB NLOS identification, but the contribution faces significant headwinds: its core ECA mechanism is a trivially transplanted 2-D image technique, its gated fusion is a well-established paradigm, and its 92.07% accuracy on eWINE does not clearly surpass classical machine-learning baselines. This audit assembles the evidence a rigorous IEEE Sensors Journal reviewer would need, organized by technical concern. The manuscript's most defensible contribution—ultra-low parameter count—is undermined by the absence of on-device deployment, quantization results, and hardware latency benchmarks that the venue demands.

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1. The eWINE accuracy landscape and where ECA-UWB sits

The eWINE LOS-NLOS dataset (Bregar, Jožef Stefan Institute, 2017) contains **42,000 samples** (21,000 LOS / 21,000 NLOS) across 7 indoor environments, GitHub collected with DecaWave DW1000 modules. GitHub +3 Each sample includes **1,016 CIR taps** ScienceDirect Taylor & Francis Online and **15 diagnostic features** (first-path amplitudes, CIR power, noise statistics, RXPACC, etc.). github GitHub A second, newer eWINE localization dataset (Bregar, 2023, *Scientific Data*) covers apartment and office environments MDPI with ~247,000 CIR samples—and some papers conflate the two, creating apples-to-oranges comparisons.

On the original 42,000-sample dataset, the accuracy trajectory is:

Method	Year	Accuracy	Params	Notes
CNN (Bregar & Mohorčič)	2018	86.45%	Large	Baseline
LSTM	2018	82.29%	Large	—
FCN-Attention (Pei et al.)	2024	88.24%	Not reported	Self-attention on FCN
CNN-CAM (Zhang et al.)	2023	90.00%	Not reported	Channel attention module
ADCNN-BiLSTM-Transformer	2025	92.19%	Large	Adaptive dilated CNN
ECA-UWB (under review)	**—**	**92.07%**	**2,374**	**This manuscript**
SA-TinyML (Wu et al.)	2024	92.92%	4,627	Self-attention TinyML
SVM (Che et al.)	2022	92.6%	Kernel-based	IEEE Sensors J.
Decision Tree (Che et al.)	2022	92.8%	Very few	IEEE Sensors J.
Naïve Bayes (Che et al.)	2022	93.2%	Minimal	IEEE Sensors J.
ResNet + Causal Backtracking	2025	96.53%	Large	Modifies dataset

Critical finding: Che et al. (IEEE Sensors Journal, 2022) reported that SVM achieves ~**92.6%** and Decision Tree ~**92.8%** University of Huddersfield on eWINE-related data—both surpassing ECA-UWB's 92.07% with negligible model complexity. If the manuscript omits these classical baselines, reviewers should flag this as a serious gap. A 2,374-parameter neural network that underperforms a decision tree is difficult to justify without demonstrating advantages in latency, generalization, or robustness that classical methods lack.

The **absence of a standardized eWINE split** is another concern. Papers use different ratios—70/10/20, 70/30, per-environment, or custom subsets—and no fixed random seed exists. The FCN-Attention paper used a 29,100/4,512/8,748 split; Taylor & Francis Online Jiang et al. (2020) subsampled to 35,000 samples with a 25,000/10,000 split. Scribd If ECA-UWB uses a 70/15/15 split, this must be scrutinized against the splits used by its direct comparators (SA-TinyML, CNN-CAM) to ensure the comparison is valid.

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2. ECA on 1-D CIR is architecturally trivial, not a contribution

ECA-Net (Wang et al., CVPR 2020; arXiv IEEE Xplore ~4,264 citations) replaces SE-Net's two fully connected layers with a **single 1-D convolution** of adaptive kernel size *k* applied along the channel dimension after global average pooling. arXiv GitHub The key insight: the ECA module is **dimensionality-agnostic**. For 2-D inputs (C×H×W), GAP produces a C×1×1 vector; for 1-D inputs (C×L), GAP produces a C×1 vector. The subsequent 1-D convolution operates identically on both. GitHub Paperspace No architectural modification is needed to move ECA from images to temporal signals.

This is confirmed by the proliferation of ECA in 1-D domains without any claimed novelty in the adaptation itself: EEG-based emotion recognition (ECA-ResDNN, *Mathematics* 2025), MDPI gearbox fault diagnosis (LECA, *Scientific Reports*), Nature PubMed Central analog circuit fault diagnosis (ISTFAN, *Electronics*), bearing fault diagnosis (MFRANet), Nature and speech emotion recognition. Each plugs ECA directly into a 1-D backbone.

More critically for this manuscript, **Niu et al. (2023, *Computers and Electronics in Agriculture*)** already applied ECA-ResNet to UWB CIR data from DW1000 modules for ranging error mitigation in greenhouses ScienceDirect (MAE = 0.1004 m, RMSE = 0.1714 m). ADS ScienceDirect While Niu's task was regression rather than classification, the architectural precedent—ECA operating on 1-D UWB CIR features—is established. Furthermore, SE attention (ECA's predecessor) has been applied directly to UWB NLOS classification: **ININN** (2022, *Digital Signal Processing*) uses SE blocks after parallel CNN+GRU branches on CIR data, ScienceDirect ResearchGate and **1D-CLANet** (Wang et al., *Applied Sciences* 2024) integrates SE attention with LSTM for multi-class UWB NLOS classification, ResearchGate achieving 95.58% accuracy on their dataset. MDPI The 1D-CLANet paper explicitly compared SE against CBAM and ECA, finding SE superior for their architecture.

Reviewer implication: Applying ECA to 1-D CIR is not a methodological contribution—it is a direct plug-in of a known module. The manuscript must demonstrate that ECA provides measurable advantages over SE and CBAM specifically on CIR data with ablation studies. If 1D-CLANet already showed ECA underperforms SE on UWB CIR, the manuscript needs to explain why its implementation differs.

3. Gated fusion is well-established, not novel for sensor data

Per-sample sigmoid-gated fusion traces directly to the **Gated Multimodal Unit (GMU)** by Arevalo et al. (ICLR Workshop 2017, expanded in *Neural Computing and Applications* 2020): $z = \sigma(W_g[x_1; x_2] + b_g)$, then $h = z \odot h_2 + (1-z) \odot h_1$. Emergent Mind +2 This paradigm has been deployed in autonomous driving sensor fusion (GRFU for CAN-bus + camera + LiDAR, Narayanan et al. 2019), multispectral object detection (GIF, Kim et al. 2018 on KITTI), Emergent Mind insurance telematics (mGMU, IEEE 2023), multimodal violence detection (MDPI 2025), and numerous other domains. Emergent Mind

In UWB NLOS specifically, the dominant fusion strategy is **concatenation + fully connected layers**. The DBFF-Transformer (Xi et al., *Information* 2025) uses exactly this architecture—a Transformer branch on CIR sequences fused with a diagnostic-feature MLP branch via concatenation— ResearchGate MDPI achieving **95.9% accuracy** on the newer eWINE localization dataset with 1.72M parameters. MDPI Si et al. (IEEE Comm. Letters 2023) similarly concatenate CNN-extracted CIR features with manual features into an MLP. IEEE Xplore IEEE Xplore LightMamba (Wang et al., IEEE IoT Journal 2026) uses dual-branch feature alignment. ResearchGate

No prior UWB NLOS paper was found using sigmoid-gated per-sample fusion between CIR and diagnostic branches. This represents application novelty—a known technique in a new domain. However, a reviewer should ask: does the gating mechanism demonstrably outperform simple concatenation? Without ablation studies comparing gated fusion vs. concatenation vs. attention-based fusion on the same backbone, the contribution reduces to "we used GMU instead of concat."

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4. Cross-dataset evaluation on VNU lacks rigor without public data

The VNU dataset appears to be a **private, unpublished dataset** likely collected at Vietnam National University. Extensive searching across academic databases, GitHub, and data repositories returned zero public references. This raises reproducibility concerns that IEEE Sensors Journal reviewers will flag.

The cross-dataset transfer literature sets a high bar. **Park et al. (Electronics, 2020)** showed CNN accuracy drops from ~98% to **~44%** when deploying across environments without adaptation—a catastrophic failure that highlights why cross-dataset evaluation matters. MDPI **Fontaine et al. (IEEE IoT Journal, 2024)** demonstrated that transfer learning with just 50 CIR samples from the target domain recovers performance to ~93% NLOS detection, using data from Ghent University's industrial testbed. University of Huddersfield **Stocker et al. (ACM, 2023)** provided the most sobering finding: cross-dataset performance "varies tremendously" due to preprocessing, labeling, and hardware differences, and blending more datasets does not necessarily help. ACM Digital Library

Standard cross-dataset protocols in the field include: (1) train on source, test on target without adaptation (to quantify degradation); (2) fine-tune on 25–400 target samples; (3) domain-adversarial training (EDANN-SOR, IEEE Trans. Instrum. Meas., 2024; IEEE Xplore AdaLOS using MMD alignment). Good cross-dataset performance without adaptation is considered **>85%**; with minimal fine-tuning, **>93%**.

Reviewer implication: If ECA-UWB merely trains on eWINE and tests on VNU (or vice versa) without reporting the degradation magnitude, the fine-tuning protocol, or the number of target samples used, this does not constitute a rigorous cross-dataset evaluation. The private nature of VNU means no reader can reproduce or verify the results.

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5. IEEE Sensors Journal demands hardware deployment for TinyML claims

IEEE Sensors Journal's reviewer guidelines require **novelty** ("new science/engineering or fresh new approaches") and **appropriateness** ("complete in and of itself"). ieee-sensors Analysis of accepted lightweight/TinyML papers in this venue reveals clear experimental expectations:

- **On-device MCU benchmarks** are strongly expected. SA-TinyML (published in IEEE IoT Journal, not Sensors Journal) evaluated on both PC and edge platforms. IEEE Xplore +3 Che et al. (IEEE Sensors Journal, 2022) and Yang et al. (IEEE Sensors Journal, 2023) included extensive real-world deployment. University of Glasgow Papers routinely target ARM Cortex-M4/M7, ESP32, or STM32H7. MDPI
- **Quantization results** (FP32 → INT8) are standard practice. SA-TinyML specifically highlighted its quantized model performance. At **2,374 parameters × 4 bytes = ~9.3 KB (FP32)** or **~2.3 KB (INT8)**, ECA-UWB's model size is genuinely impressive—but this must be demonstrated, not merely calculated.
- **Memory footprint** (Flash, SRAM), **inference latency** (ms on MCU), and ideally **power consumption** (μJ/inference) are expected differentiators.
- **Non-neural baselines** (SVM, Random Forest, XGBoost, LightGBM) are essentially mandatory. Every UWB NLOS paper published in IEEE Sensors Journal includes them. DNB

If ECA-UWB reports only PC-based accuracy and parameter counts without MCU deployment, quantization, or classical ML baselines, it falls below the venue's experimental standard. The "lightweight" framing becomes aspirational rather than demonstrated.

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6. The parameter-accuracy trade-off is defensible but narrow

The core value proposition—**49% parameter reduction** (4,627 → 2,374) for **0.85 percentage point** accuracy loss (92.92% → 92.07%)—is within accepted TinyML bounds. The Lottery Ticket Hypothesis literature shows ResNet-18 subnetworks at 10–20% original size losing only ~0.9pp on CIFAR-10. Mlsysbook INT8 quantization typically costs 0.5–2pp. NVIDIA Developer MDPI MobileNet-family architectures explicitly trade accuracy for efficiency.

ECA-UWB's parameter efficiency ratio (**0.0388% accuracy per parameter**) is ~1.93× that of SA-TinyML (0.0201%). This is a real metric. However, several factors weaken the argument:

LightMamba (Wang et al., IEEE IoT Journal, 2026) achieves F1 **95.20%** on the OIUD dataset with 22k parameters and millisecond-level inference. ResearchGate While it has ~9× more parameters than ECA-UWB, its accuracy advantage may be large enough to dominate on the Pareto frontier when considering practical deployment. Its eWINE-specific accuracy was not publicly available but it was evaluated on eWINE, IULCD, and OIUD—three datasets ResearchGate versus ECA-UWB's two (eWINE + private VNU). ResearchGate

MS-CNN-SA (Jiang et al., IEEE Comm. Letters, 2026) reportedly targets MCU deployment with multi-scale CNN and self-attention, but could not be located in databases—it may be in early access or the citation may be imprecise.

DBFF-Transformer achieves 95.9% accuracy with a dual-branch architecture conceptually similar to ECA-UWB (CIR branch + diagnostic feature branch), though at 1.72M parameters it targets a different complexity regime. MDPI ResearchGate

The most damaging comparison remains **classical ML**. If SVM achieves 92.6% on the same dataset with a model that fits in bytes rather than kilobytes, the neural network's advantage must be articulated in terms beyond raw accuracy— University of Huddersfieldperhaps in latency consistency, batch inference throughput, or graceful degradation under distribution shift. Without this argument, reviewers will question why the problem needs a neural network at all.

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7. Specific reviewer concerns and missing elements

Based on this audit, an IEEE Sensors Journal reviewer should raise the following structured concerns:

Novelty concerns. The manuscript combines three established components—ECA (CVPR 2020, trivially adapted to 1-D), Semantic Scholar arXiv gated fusion (GMU, 2017), and dual-branch CIR+diagnostic architecture (DBFF-Transformer 2025, ResearchGate Si et al. 2023, IEEE Xplore LightMamba 2026)—without clear evidence that the combination produces emergent benefits beyond what each component contributes individually. Niu et al. (2023) already used ECA on UWB CIR. ResearchGate +2 The 1D-CLANet paper (2024) compared ECA against SE and CBAM on UWB signals and found SE superior. ResearchGate The paper must explain why ECA is the right choice for this task against this existing evidence.

Missing baselines. SVM (~92.6%), Decision Tree (~92.8%), and Naïve Bayes (~93.2%) from Che et al. (IEEE Sensors Journal 2022) all match or exceed ECA-UWB's 92.07%. University of Huddersfield LightGBM and XGBoost baselines are absent. Springer The DBFF-Transformer's dual-branch architecture is a direct structural comparator. MDPI ResearchGate LightMamba's Mamba-based approach with dual-branch extraction is the most recent lightweight competitor. ResearchGate

Evaluation protocol gaps. The eWINE dataset lacks a standardized split; the manuscript must specify its exact split and justify comparability with SA-TinyML's evaluation. Two distinct eWINE datasets exist (the original 42K-sample LOS-NLOS set and the newer ~247K-sample localization set)— ScienceDirect +3 the manuscript must clarify which is used. GitHub The VNU cross-dataset is private and unverifiable. Per-class metrics (NLOS precision/recall, F1) should be reported alongside accuracy, as NLOS detection on eWINE is notably harder than LOS (NLOS specificity ~84.9% vs. LOS sensitivity ~91.6% in FCN-Attention). Taylor & Francis Online

Hardware deployment absence. No MCU benchmarks, no quantization results, no latency measurements, no memory footprint analysis. Carloalbertoboano For a paper framing itself around "lightweight" design with 2,374 parameters, the absence of deployment evidence is a critical weakness for IEEE Sensors Journal.

Ablation insufficiency. Reviewers should request: (1) ECA vs. SE vs. CBAM vs. no attention on the CIR branch; (2) gated fusion vs. concatenation vs. addition vs. attention-weighted fusion; (3) dual-branch vs. CIR-only vs. diagnostic-only; (4) sensitivity to CIR segment length (50 vs. 120 vs. full 1,016 taps). arXiv

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Conclusion

ECA-UWB's **2,374-parameter footprint** is genuinely small and represents a legitimate contribution to the ultra-constrained deployment space—if validated on hardware. The 0.85pp accuracy trade-off versus SA-TinyML is acceptable by TinyML standards. However, the manuscript faces a compounding novelty problem: each individual component (ECA on 1-D signals, GitHub sigmoid-gated fusion, dual-branch CIR+diagnostic architecture) has clear, published precedent, and the assembly does not clearly produce emergent value. The absence of MCU deployment, quantization, classical ML baselines, and a reproducible cross-dataset protocol places the work below IEEE Sensors Journal's experimental standards as currently described. The strongest path to acceptance would require: (1) on-device MCU deployment with measured latency, memory, and power; (2) INT8 quantization results; (3) SVM/XGBoost/LightGBM baselines; (4) comprehensive ablations; and (5) a clear articulation of the deployment scenario where 2,374 parameters enables functionality that 4,627 parameters cannot.